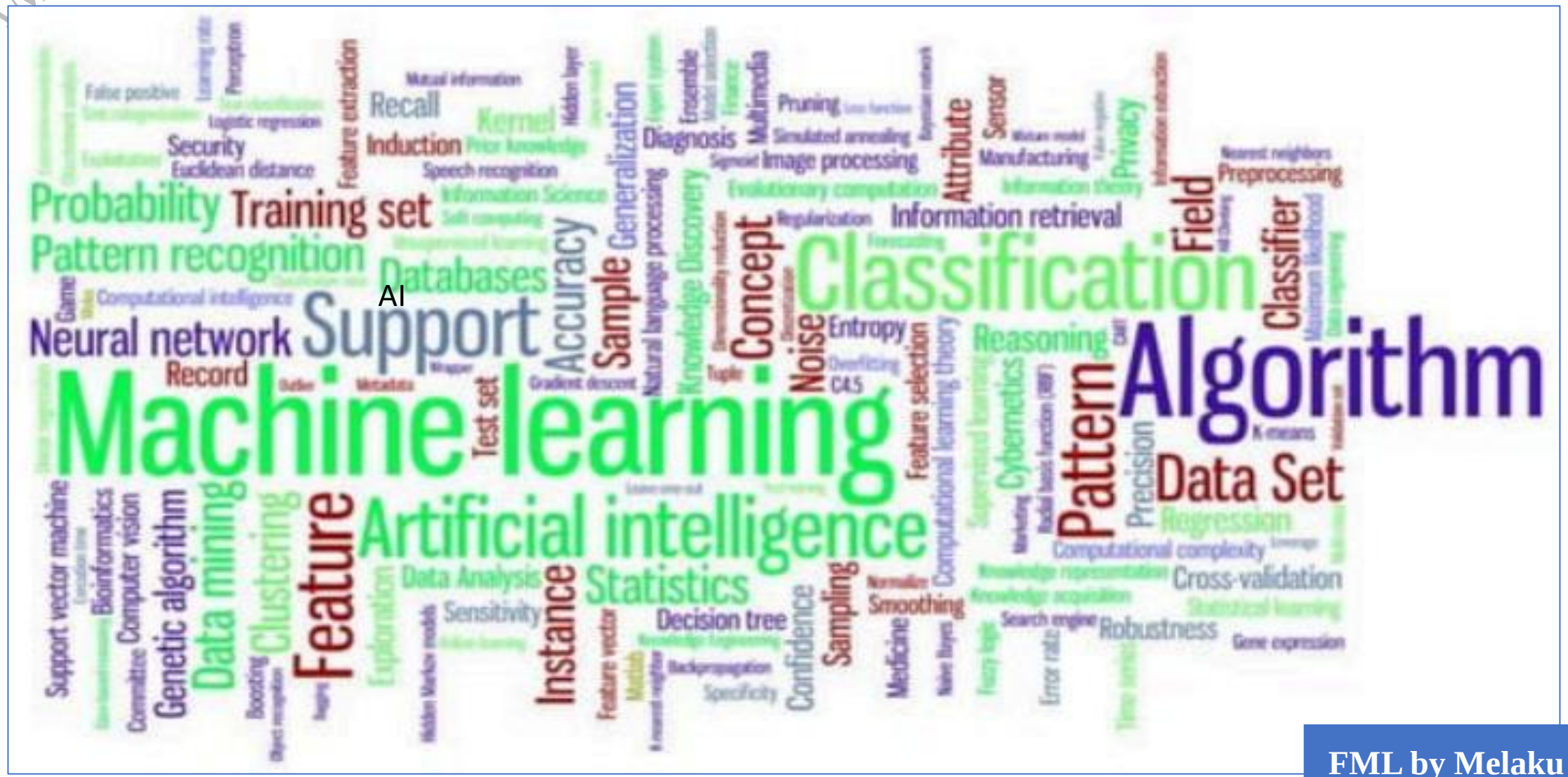
Fundamentals of Machine Learning

Instructor: Melaku M.

Target Group: G3SEng

AMU - FCSE

Chapter 1: Machine Learning Fundamentals



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Quotes

❖"If you were a current computer science student, what area would you start studying heavily?

- Answer: Machine Learning.

–Bill Gates, Reddit AMA

❖"Machine learning is today's discontinuity"

–Jerry Yang, Co-founder, Yahoo

❖"AI is the new electricity! Electricity transformed countless industries; AI will now do the same." –Andrew Ng

What is Machine Learning?

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Potential Definitions for Machine Learning

❖Machine learning (ML) is a field of study that focuses on creating systems that can learn from data without being explicitly programmed. “Arthur Samuel (1959)”

❖ Essentially, it allows computers to:

- **Learn from data:** Instead of relying on rigid, pre-written/hard-coded rules, ML models train on large datasets to recognize patterns and relationships.
- **Improve performance over time:** As they are exposed to more data, ML models refine their understanding and become more accurate in their predictions or decisions.
- **Make predictions or decisions**: Based on the patterns they've learned, ML algorithms can predict future outcomes or make informed decisions.

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Potential Definitions for Machine Learning

Definition by **Tom Mitchell** (1998):

Machine Learning is the study of algorithms that:

- improve their performance *P*
- at some task *T*
- with experience *E*

A well−defined learning task is given by <*P*, *T*, *E*>

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Defining the Learning Task

- Improve on task T, with respect to performance metric P, based on experience E

T: Recognizing and classifying hand-written words within images

P: Percentage of words correctly classified

E: Dataset of human-labeled images of handwritten words

T: Classify emails as legitimate or spam

P: Percentage of emails labeled correctly

E: Repository of emails, some with human-specified labels

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Traditional Programming Vs Machine Learning

Traditional Software Development



- **Static** – hard-coded set of steps and scenarios
- **Rule Based** – expert knowledge
- **No generalization** – handling special cases is difficult

Machine Learning Programming



- **Dynamic** – evolves with data, finds new patterns
- **Data driven** – discovers knowledge
- **Generalization** – adapts to new situations and special cases

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When We Need Machine Learning

- Tasks requiring customization



  - Email filters
  - Personalized news
  - Personalized Tutoring

Tasks that adapt and customize themselves to individual users.

- Tasks involving big data



  - Genomics
  - Internet search
  - Anomaly detection

Discover new knowledge from large databases

- Tasks for which we don't have human expertise



  - Space exploration
  - Undersea manipulation
  - Cellular robotics

- Tasks for which it is challenging to specify our knowledge



  - Facial recognition
  - Understanding speech
  - Medical diagnosis

Too difficult/expensive to construct manually

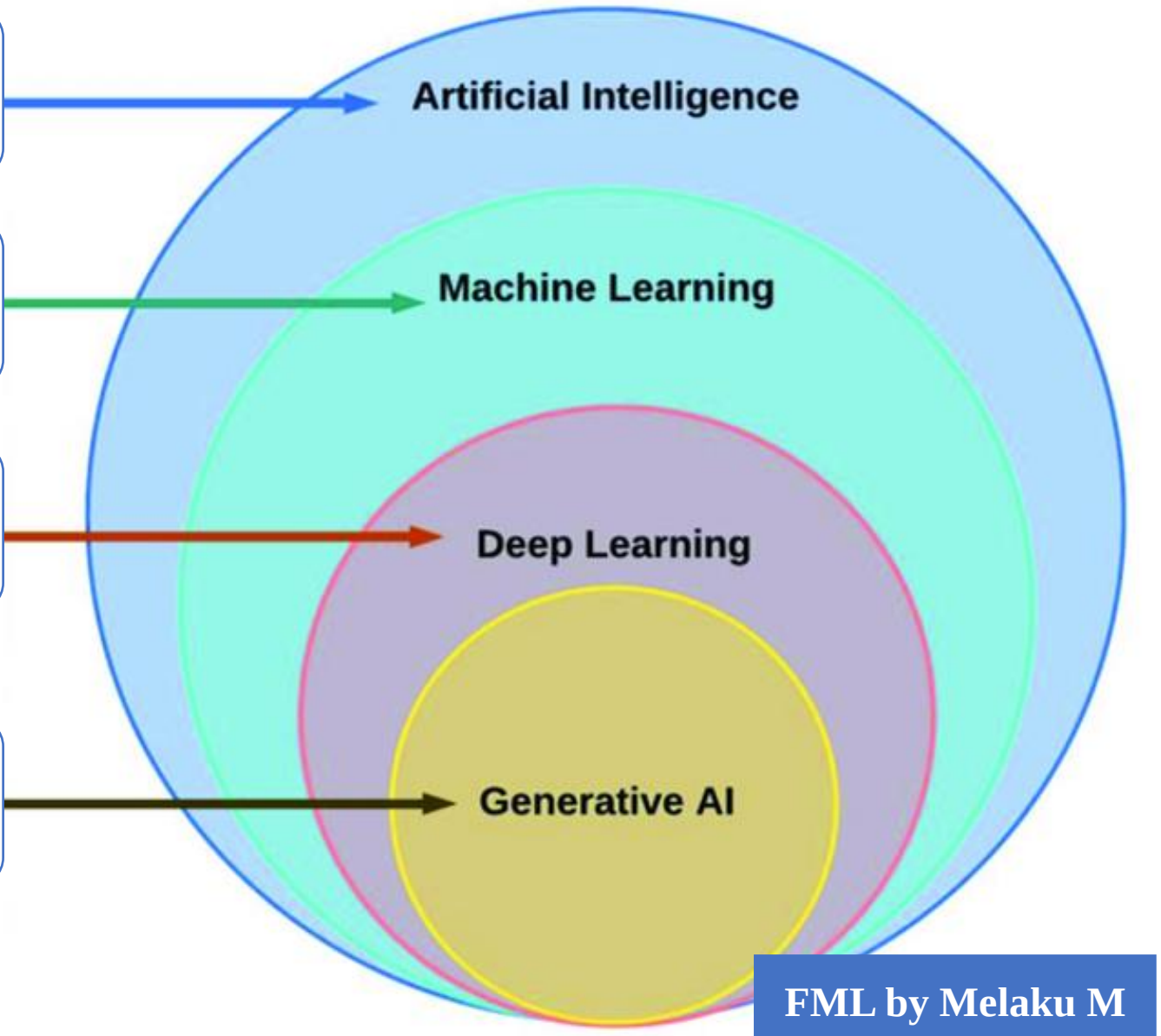
Demystifying AI, ML, Deep Learning and Generative AI

Ability of a machine to imitate intelligent human behavior. Broad (any intelligent system)

Gives the computers "**the ability to learn** without being explicitly programmed.

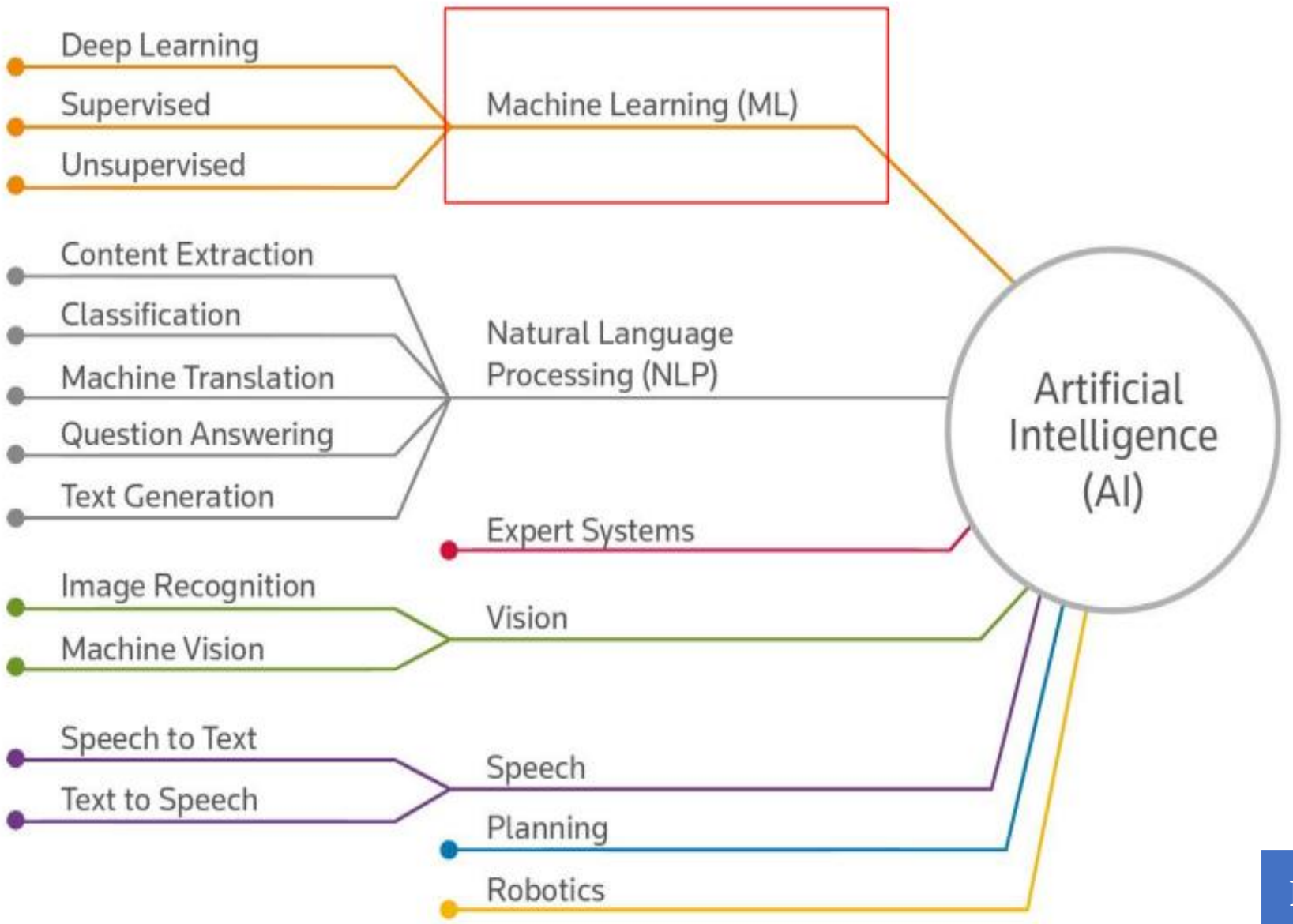
Uses neural networks with many layers (hence "deep") to learn complex patterns from large amounts of data.

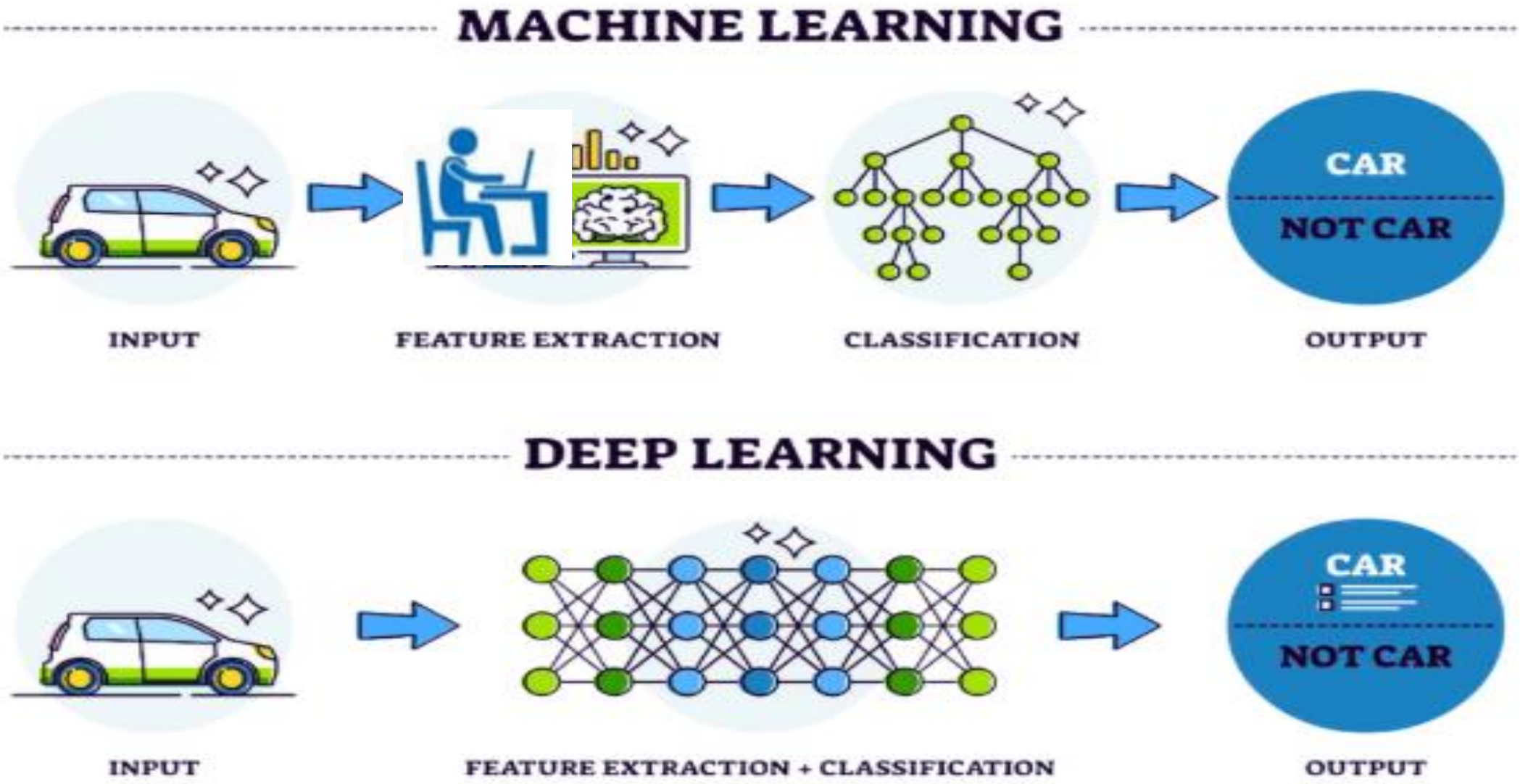
Focused on **creating content**, such as text, images, music, or code, by learning patterns from existing data.



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Artificial intelligence (AI)-Broader concept





DL: Automatically extracts features from raw data, reducing the need for manual feature en

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State-of-the-Art Applications of Machine Learning

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Generative AI



DALL-E

ChatGPT

GitHub Copilot



Figure: generative AI platforms

•**ChatGPT:** This LLMs has a foundation of GPT architecture that generates text that resembles something a human would produce. It's a helpful companion for research, strategy, and content creation.

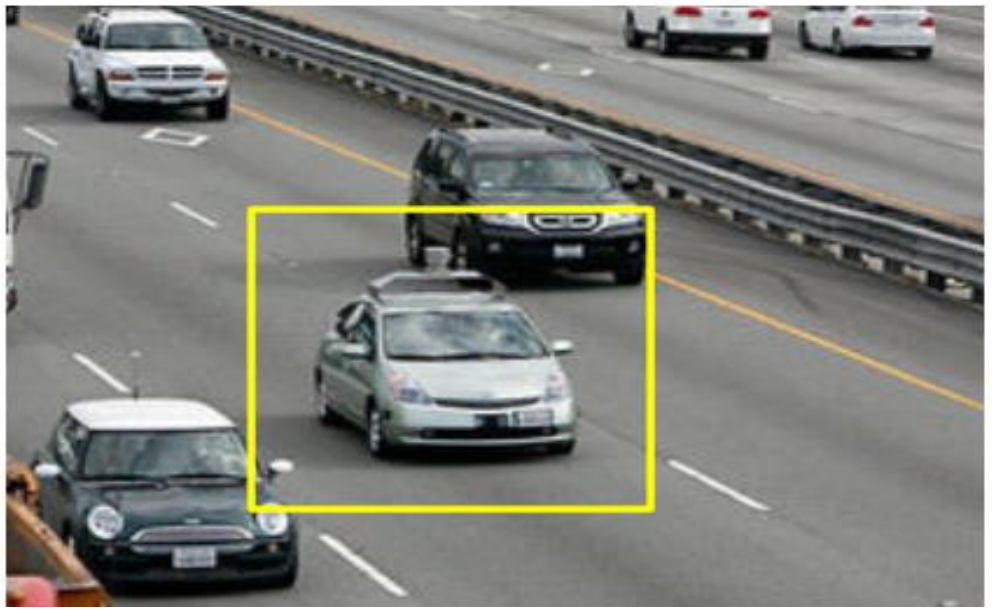
•**DALL-E2:** This model generates images from text prompts, so creatives can create vibrant illustrations and concept art that's a useful accompaniment to content marketing.

•**GitHub Copilot:** This collaboration between GitHub and OpenAI acts as a coding companion to help developers code faster and more intuitively.

•**Gemini:** large language model chatbot, also known as a conversational AI .

Autonomous Cars





- Nevada made it legal for autonomous cars to drive on roads in June 2011 .
- As of 2019, 37 states have enacted legislation regarding autonomous cars.

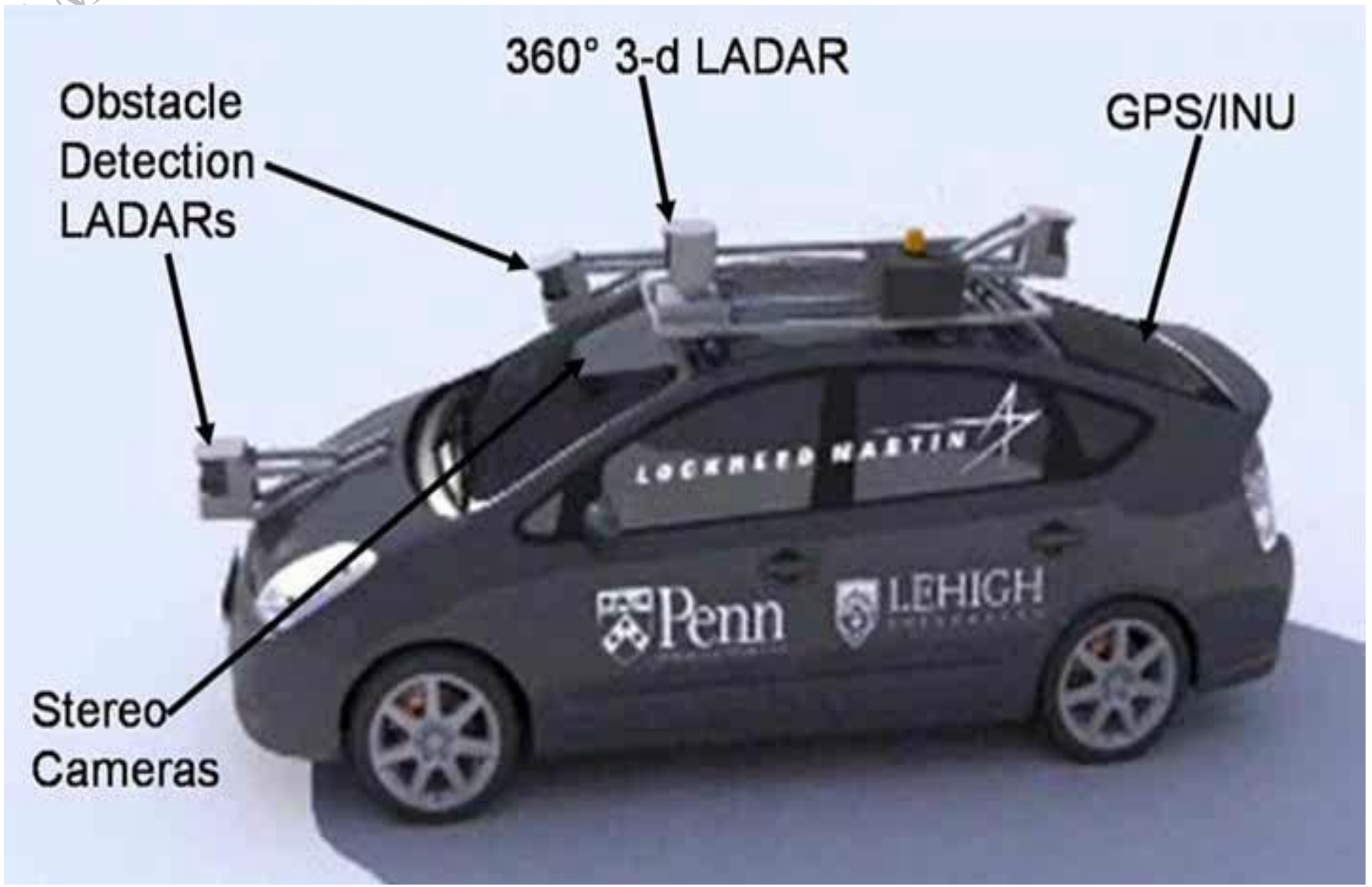


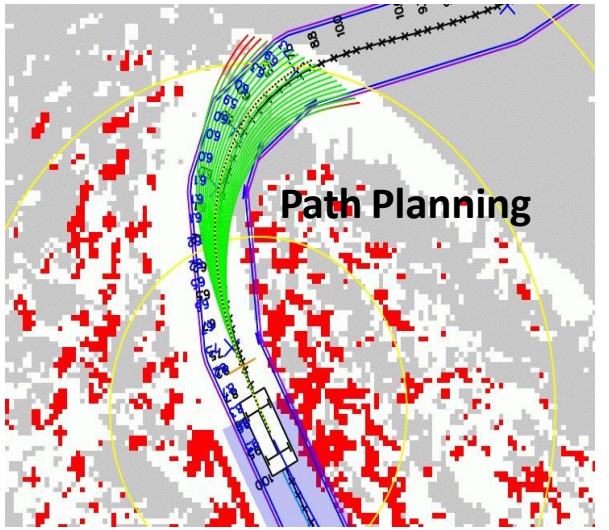
Penn's Autonomous Car à (Ben Franklin Racing Team)

FML by Melaku M

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Autonomous Cars







Deep Learning in the Headlines



BloombergBusinessweek
Technology

Acquisitions

The Race to Buy the Human Brains Behind Deep Learning Machines

By Ashlee Vance | January 27, 2014

intelligence projects. "DeepMind is bona fide in terms of its research capabilities and depth," says Peter Lee, who heads Microsoft Research.

According to Lee, Microsoft, Facebook (FB), and Google find themselves in a battle for deep learning talent. Microsoft has gone from four full-time deep learning experts to 70 in the past three years. "We would have more if the talent was there to

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Deep Learning's Role in the Age of Robots

BY JULIAN GREEN, JETPAC 05.02.14 2:56 PM

CNN BUSINESS Markets Tech Media Success Perspectives Videos

BUSINESS EVOLVED

How elite investors use artificial intelligence and machine learning to gain an edge

By Matt Egan, CNN Business
Updated 11:14 AM ET, Sun February 17, 2019

Google has started using AI to build more advanced AI

Follow @BINordic | Follow @BINordic | 2,708 followers

David Nield | ScienceAlert | 22 May 2017 11:44 AM | 1227

INDEPENDENT

NEWS VOICES SPORT CULTURE INDY/LIFE INDYBEST VIDEO DAILY EDITION COUPONS

Rise of the Machines: Can AI Beat Humanity at its Own Game?

"I believe there is no deep difference between what can be achieved by a biological brain and what can be achieved by a computer. It, therefore, follows that computers can, in theory, emulate human intelligence—and exceed it" — Steven Hawking

By Guest Writer - December 26, 2018 701





Google creates AI that can teach itself to be better than humans

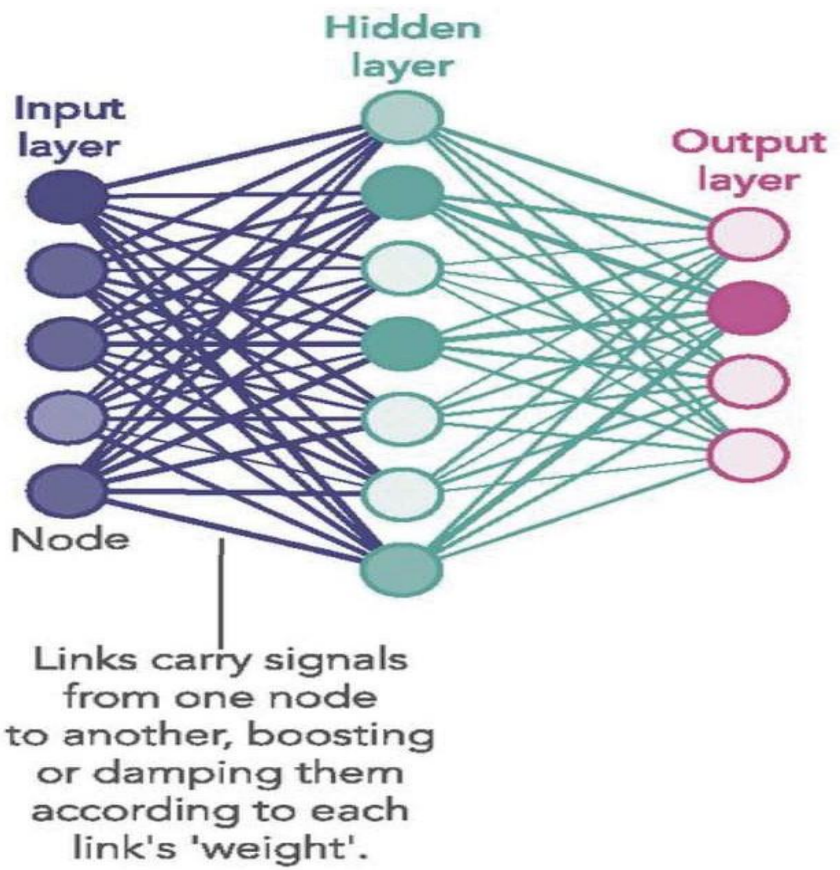
It became one of the greatest Go players of all time in a matter of days, when humans have been playing the game for thousands of years

Bloombe Technology

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Deep Networks Learn Layered Representations

1980s-Era Neural Network



Deep Neural Networks

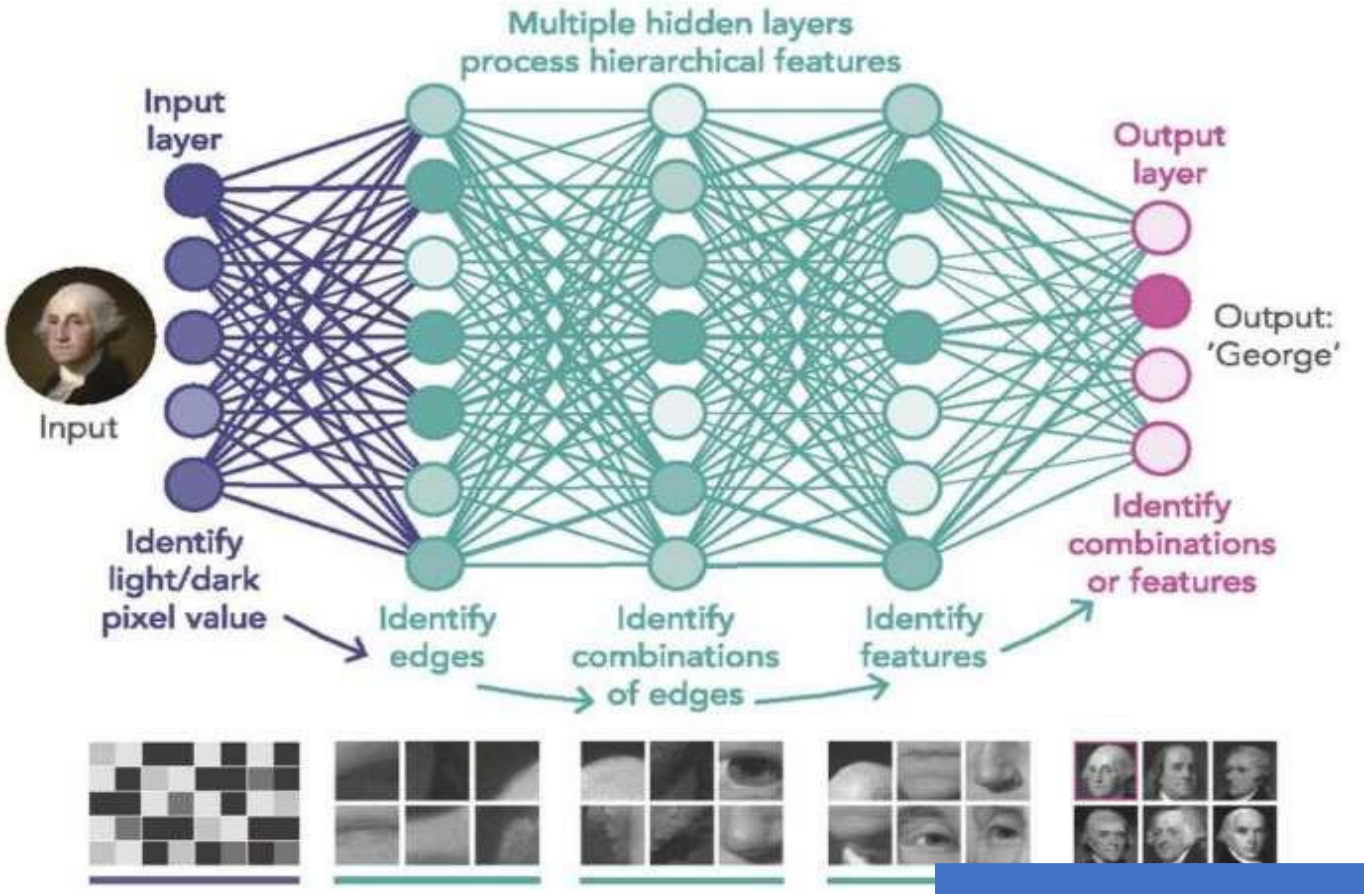
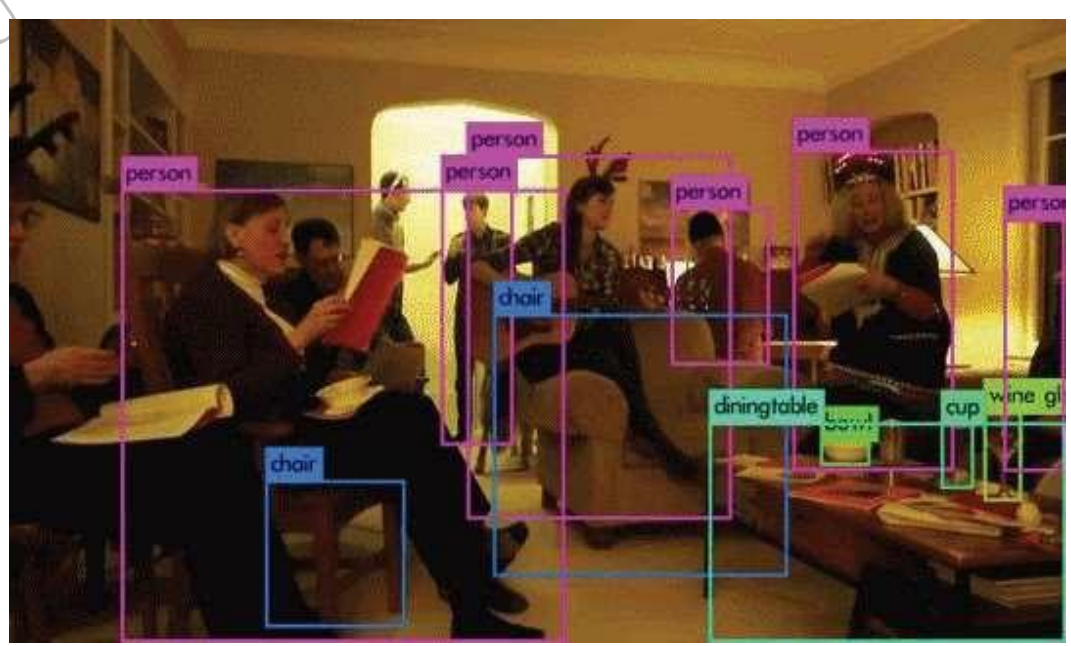


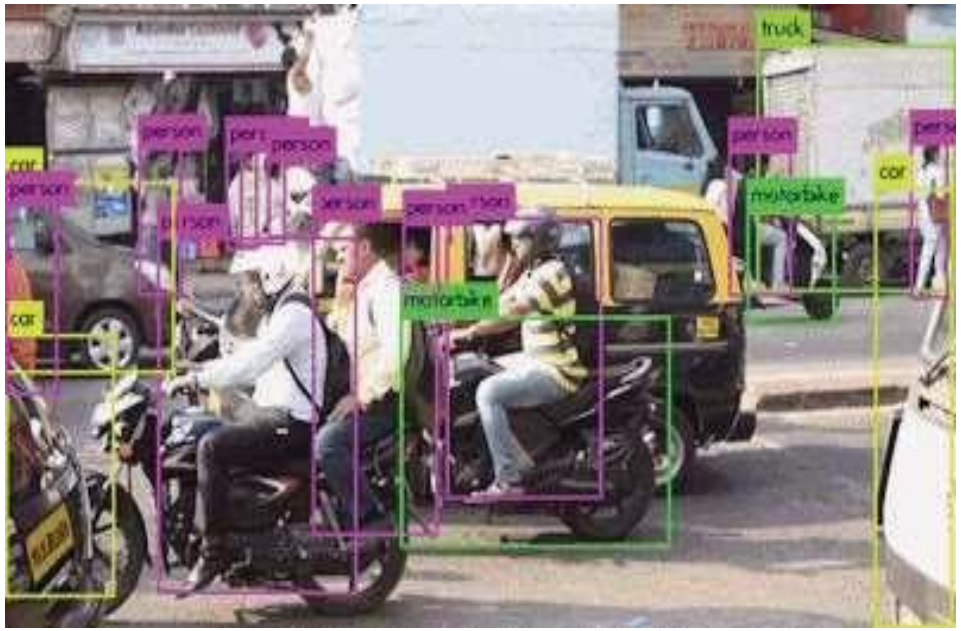
Image: https://www.pnas.org/con

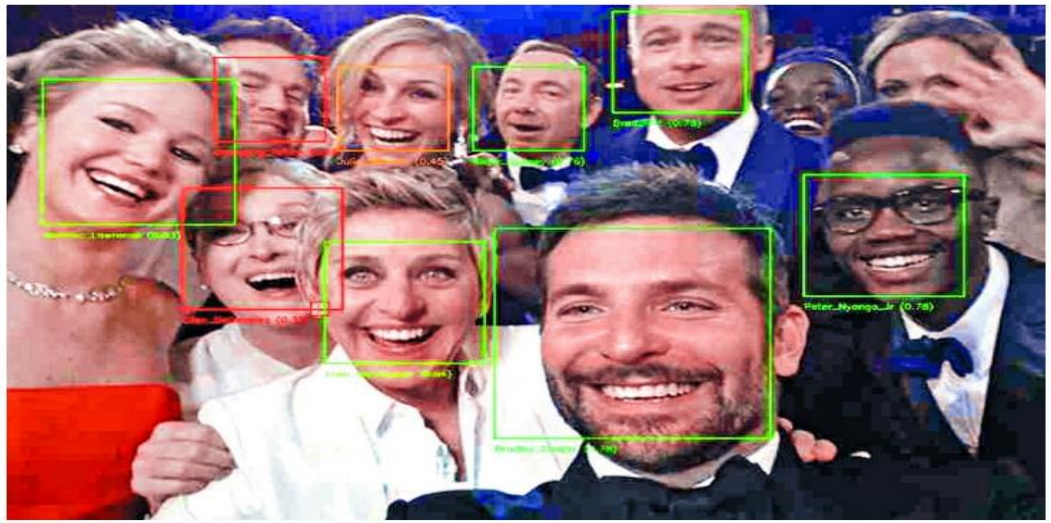
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Object Recognition

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Image Translation- Sketch to Photo



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Image Synthesis – Image Inpainting





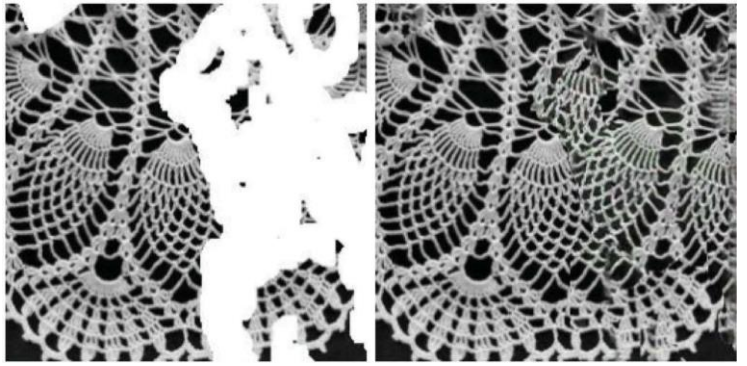




Image inpainting is essentially the art of filling in missing or damaged parts of an image.

E.g., Repair scratches, cracks, Take out unwanted objects, Fill in areas for artist

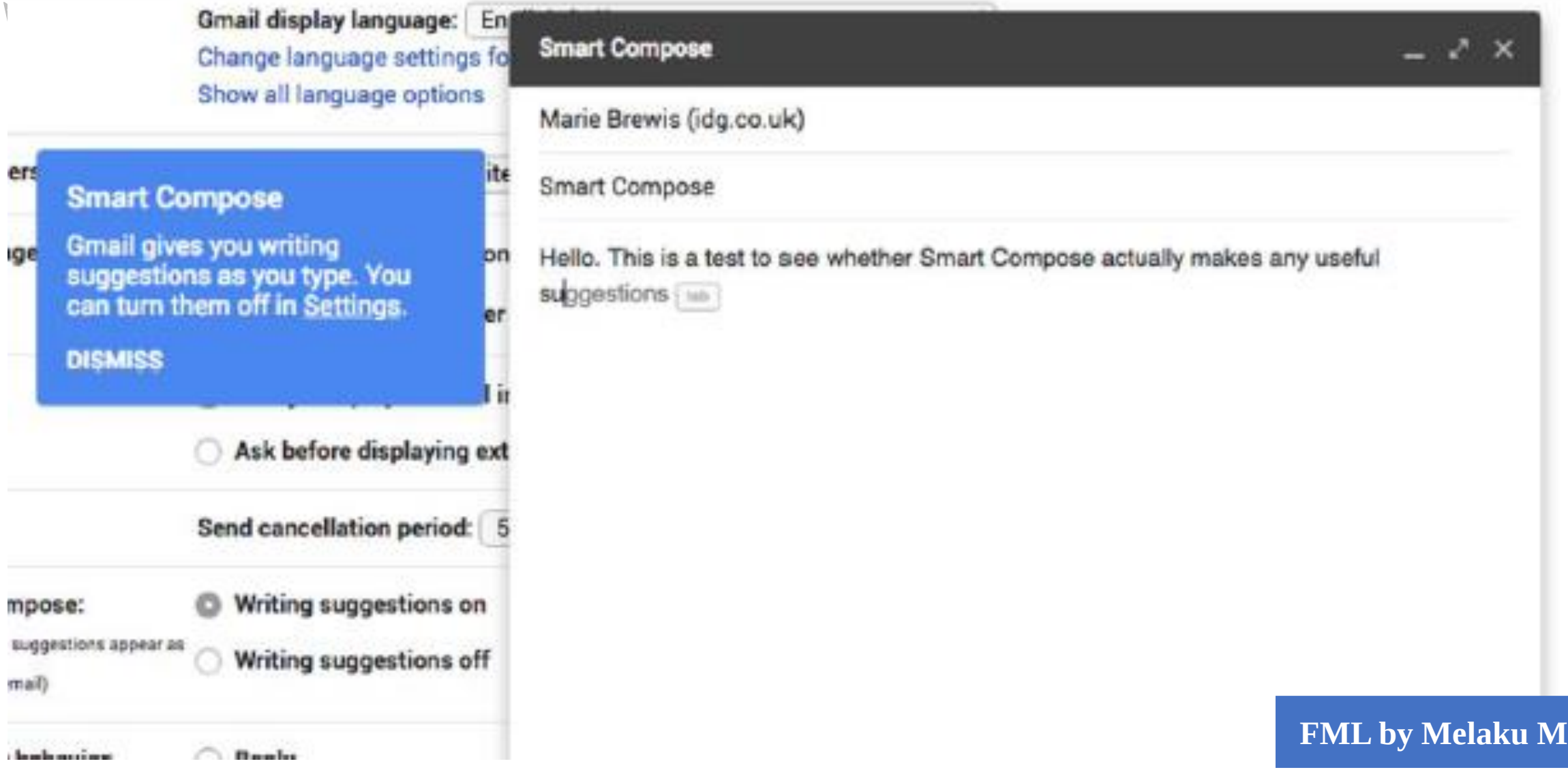
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NLP-Named Entity Recognition



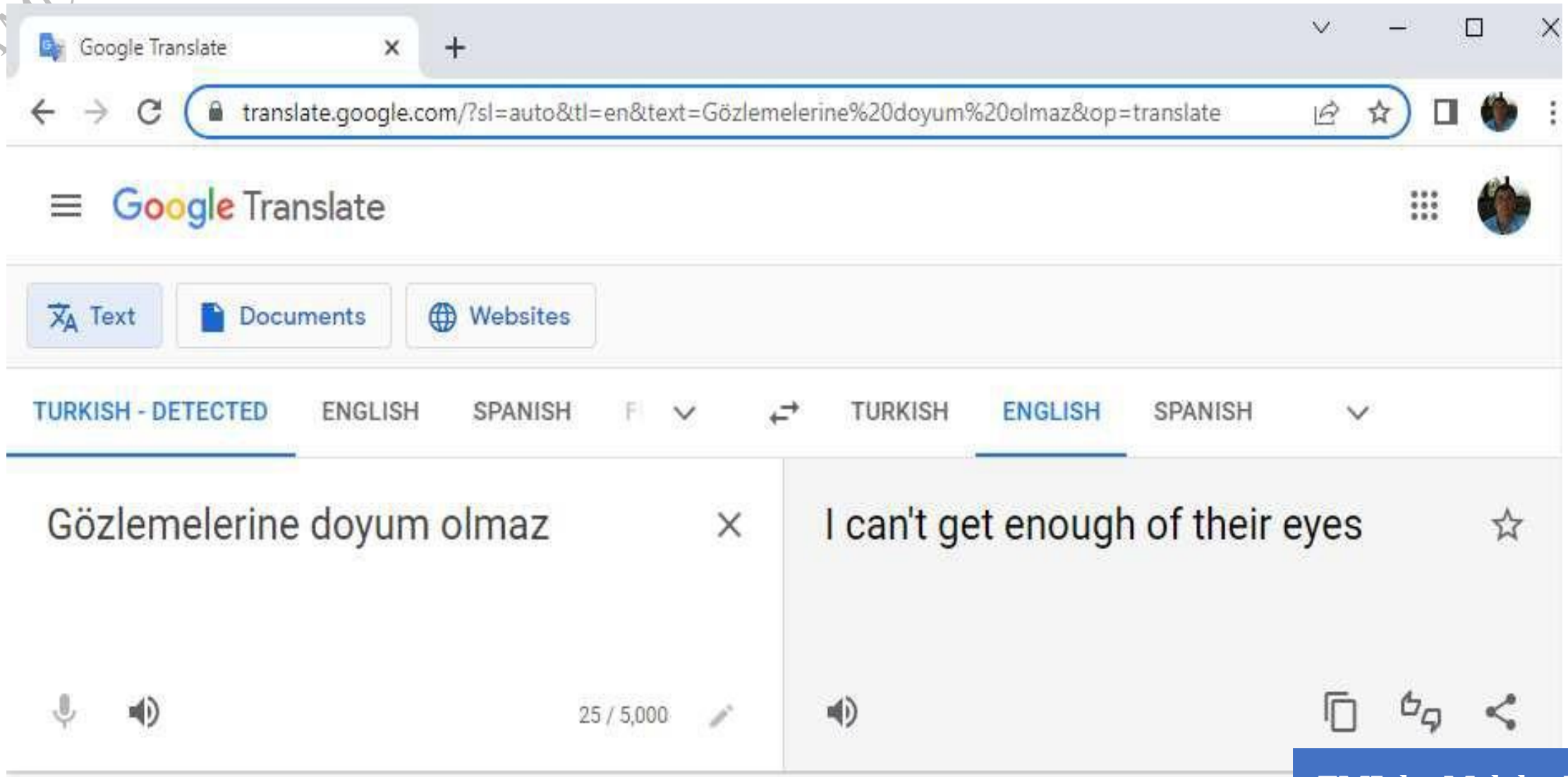
NLP: Text Generation



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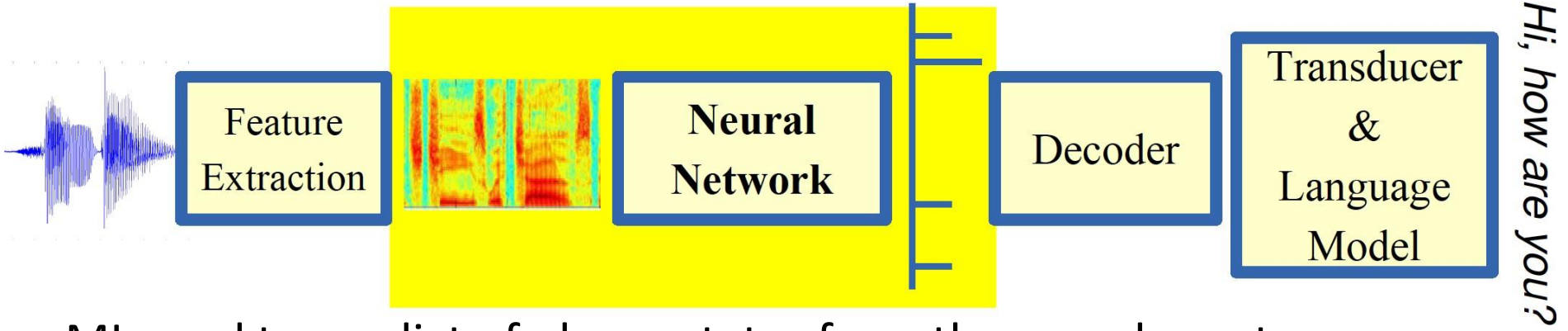
NLP: Text Translation/Machine Translation



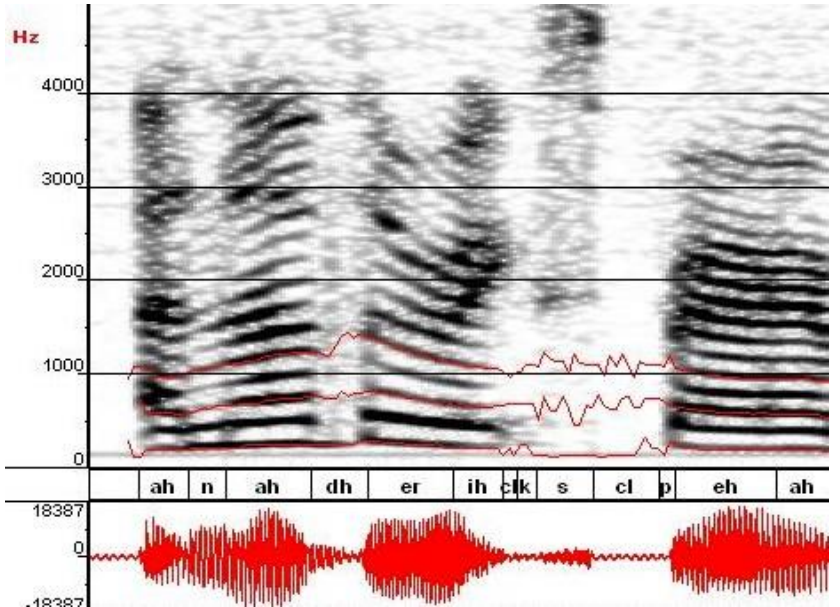
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Automatic Speech Recognition

A Typical Speech Recognition System



ML used to predict of phone states from the sound spectrogram



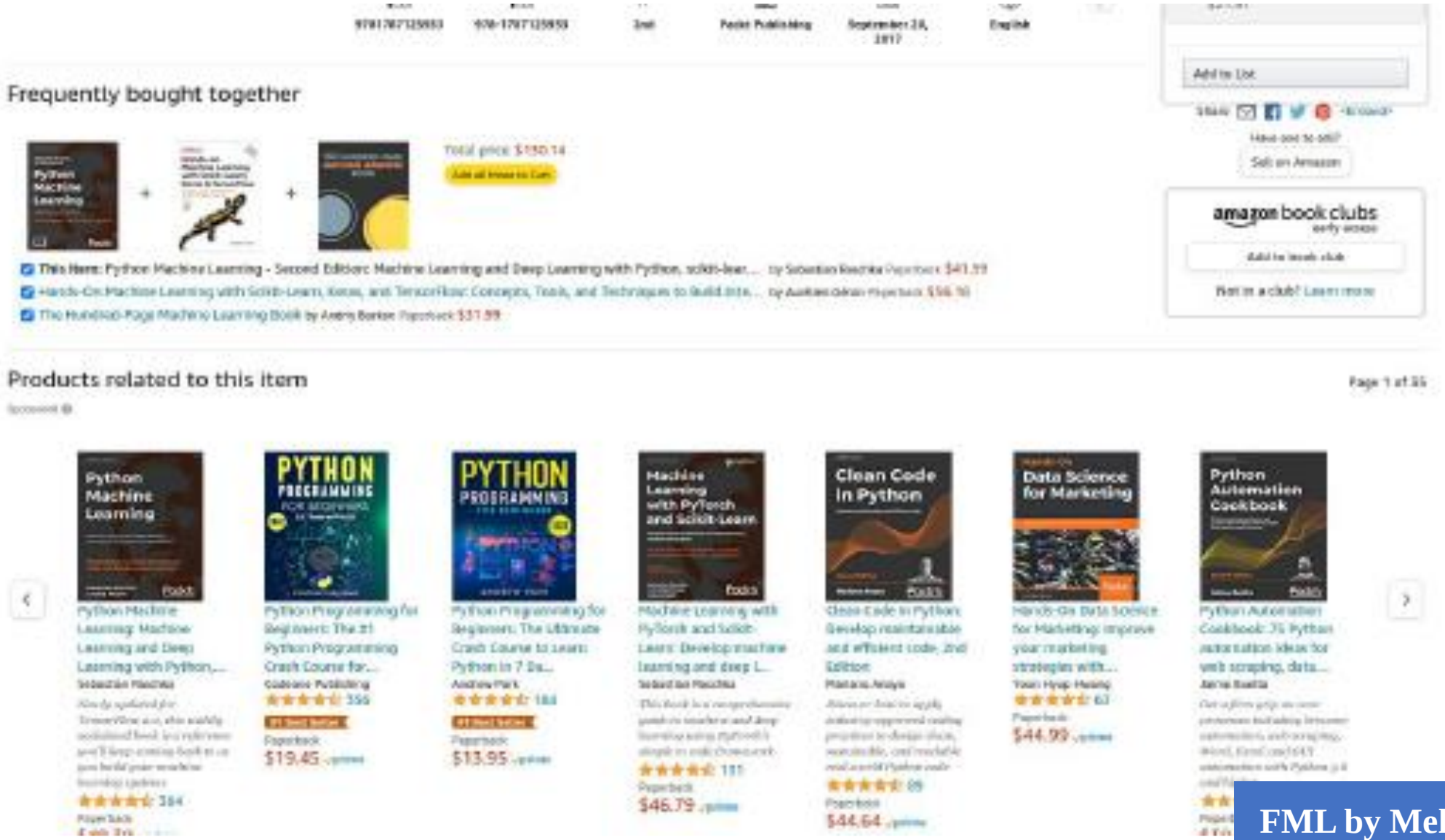
Deep learning has state-of-the-art results

# Hidden Layers	1	2	4	8	10	12
Word Error Rate %	16.0	12.8	11.4	10.9	11.0	11.1

Baseline GMM performance = 15.4%

[Zeiler et al. “On rectified linear units for speech recognition” ICASSP 2013]

Recommendation Systems



FML by Melaku M

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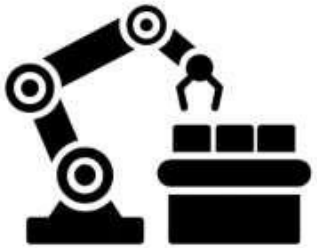
Machine learning is currently the preferred approach in the following domains:



Web Search



Computational Biology



Robotics



Finance



Social Networks



Software Debugging



Space Exploration



E-Commerce



Information Extraction



Your Favorite

Machine Learning pipeline

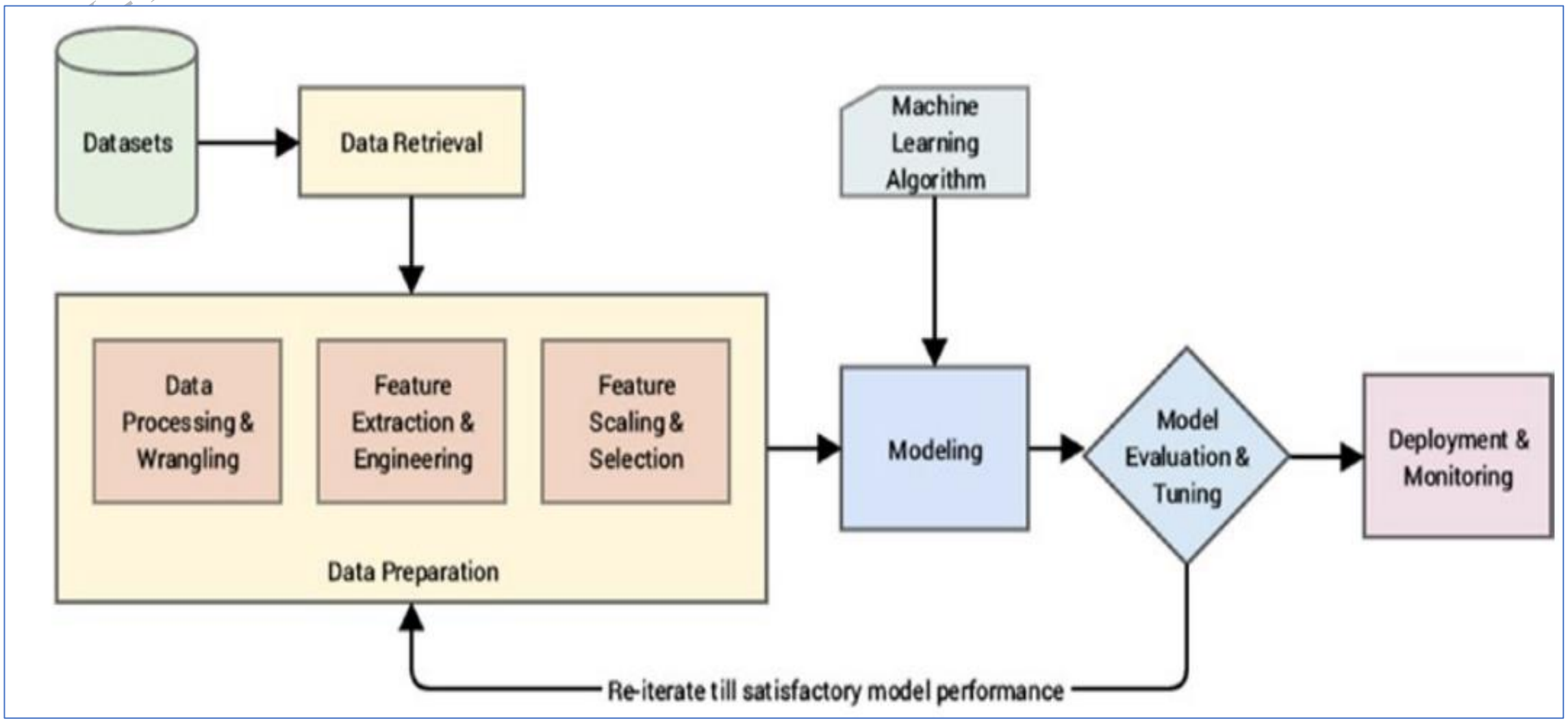
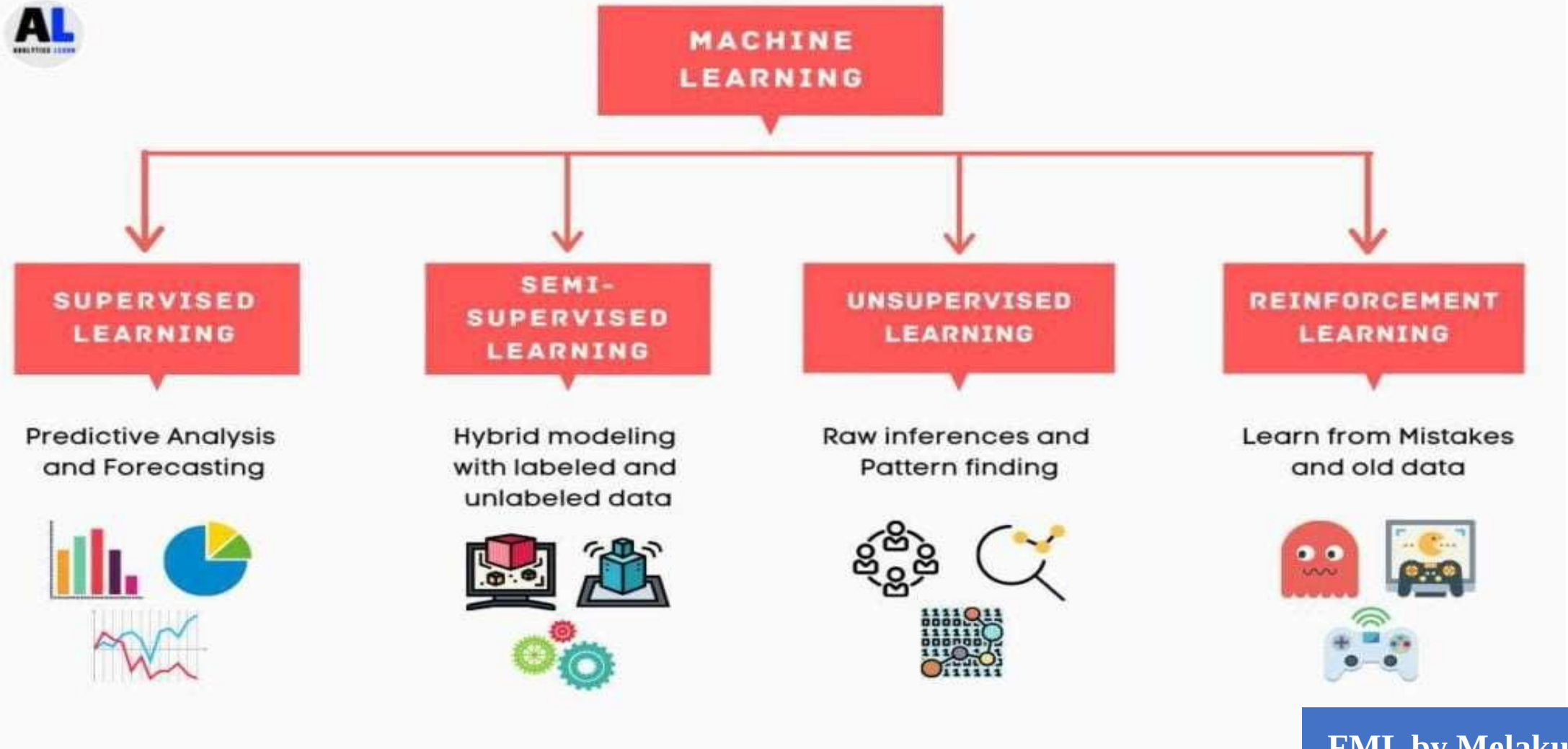
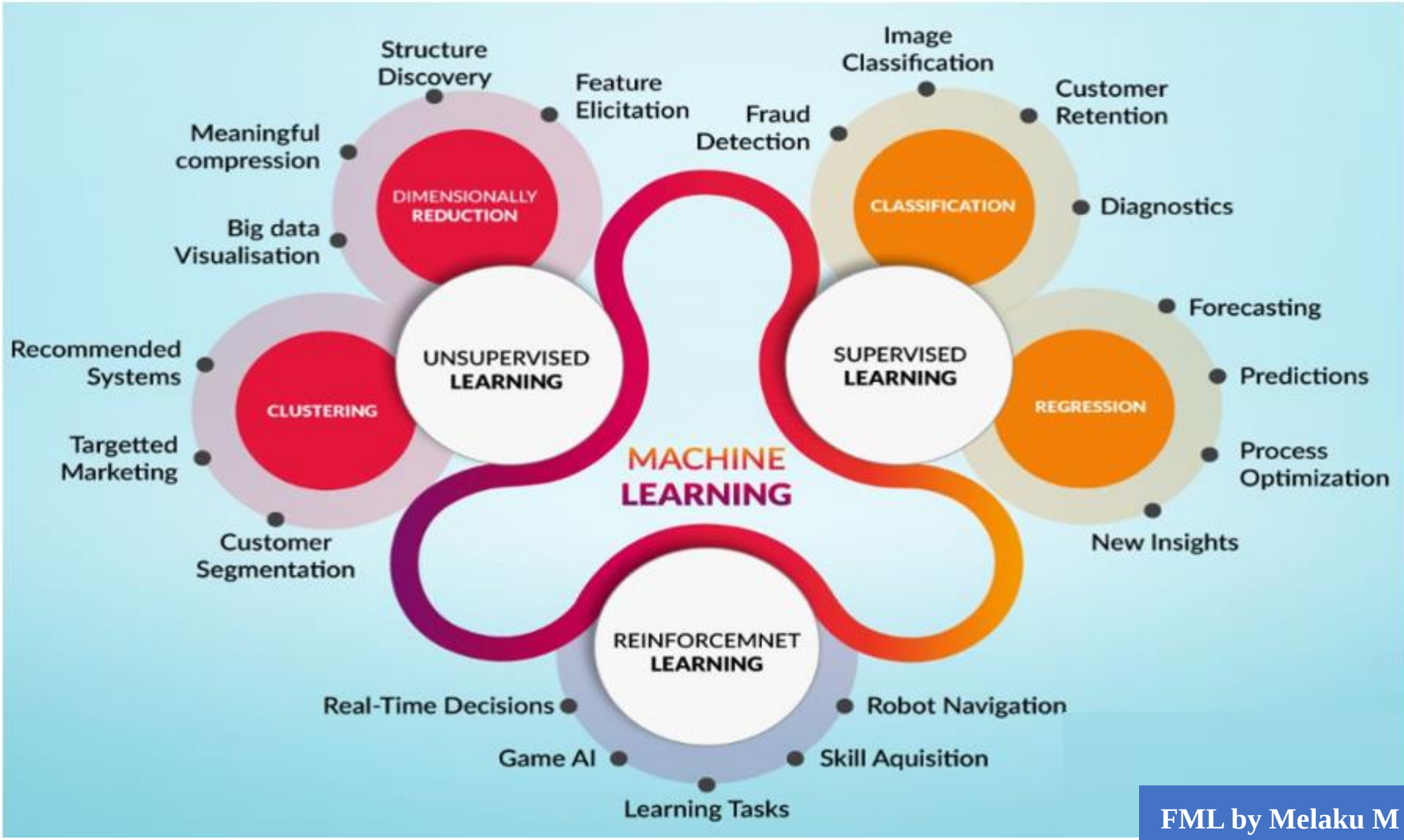


Figure: A standard Machine Learning pipeline

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Types of Machine Learning





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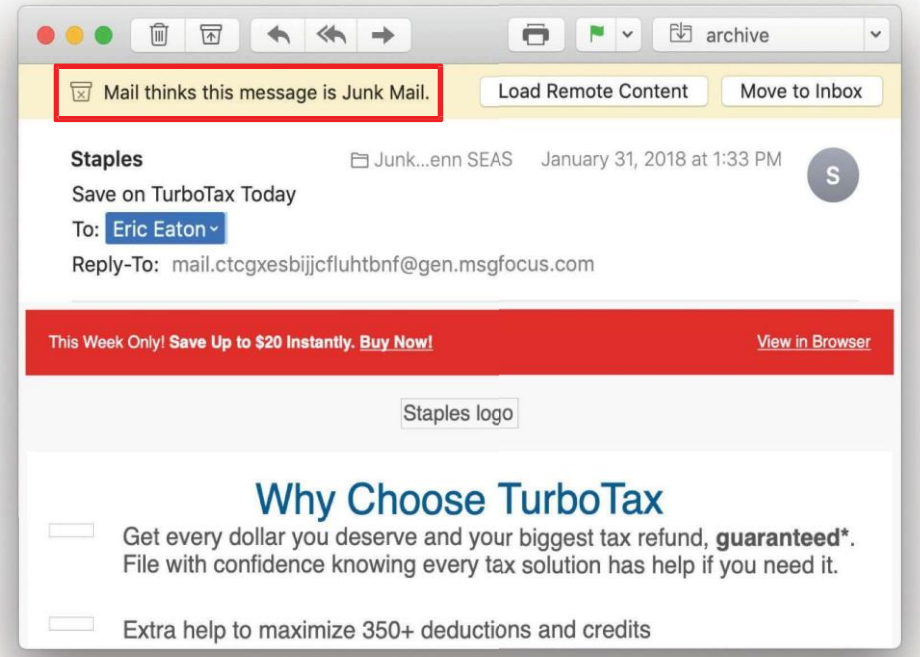
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Supervised Learning

- The model is trained on labeled data, meaning that the input data is paired with the correct output.
- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
  - Learn a function $f(x)$ to predict y given x
  - y is categorical == classification
  - y is real-valued == regression

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Supervised Learning: Spam Detection



- This is a binary classification task: Assign label (i.e. spam/not-spam) to the input (an email message)
- Classification requires a model (a classifier) to determine which label to assign to input.

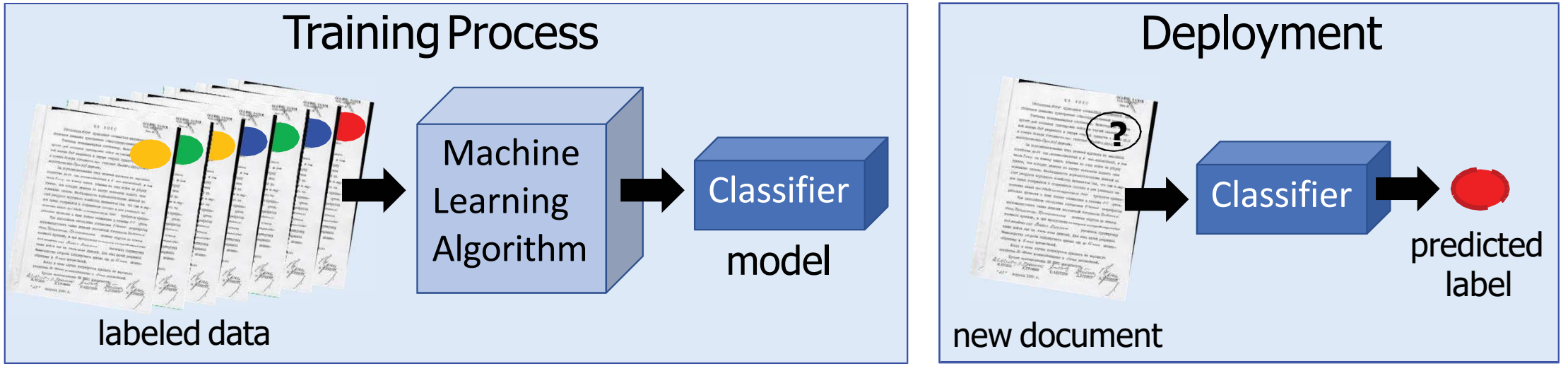


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Supervised Learning: Document Classification

- This is a multi-class classification task:

  Assign label (i.e.Politics, Sports, Finance, Arts) to the input



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Supervised Learning: Document Classification

- This idea generalizes to many types of data and applications

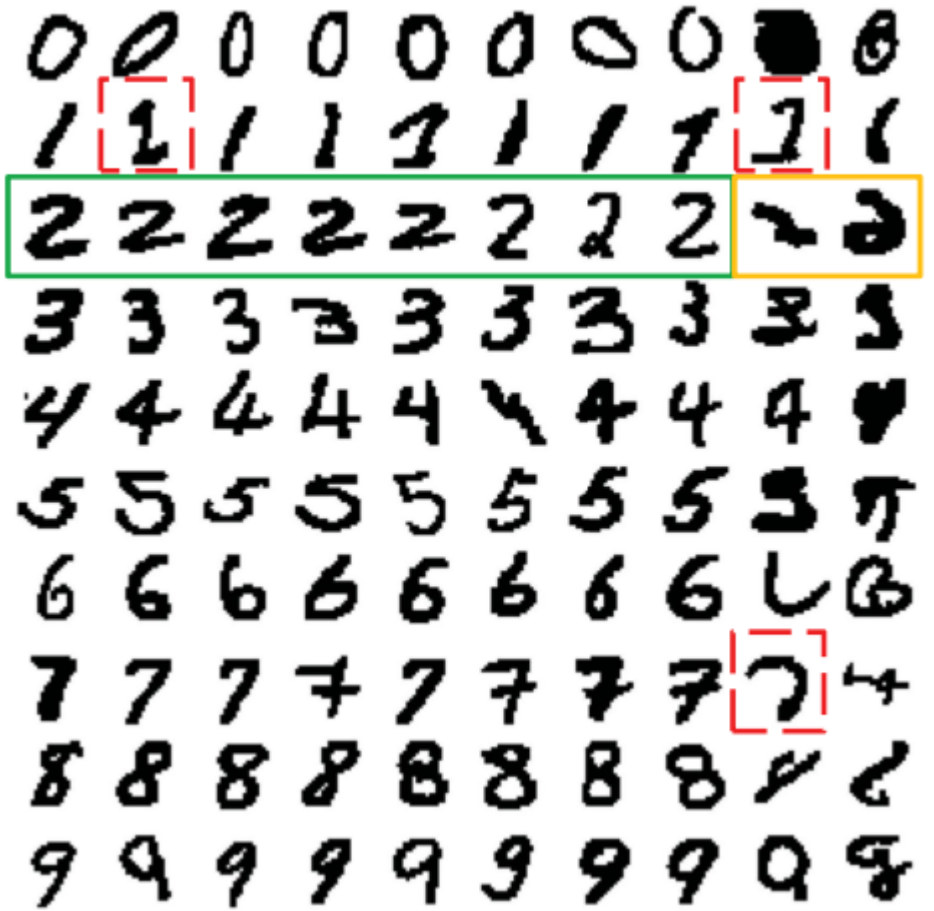


Data	Labels
Documents	Politics,Sports,Finance
Sentences	Positive,Negative
Phrases	Person,Location
Images	Cat,Dog,Snake,Horse Re-
M edical records	admit soon/Not
…	

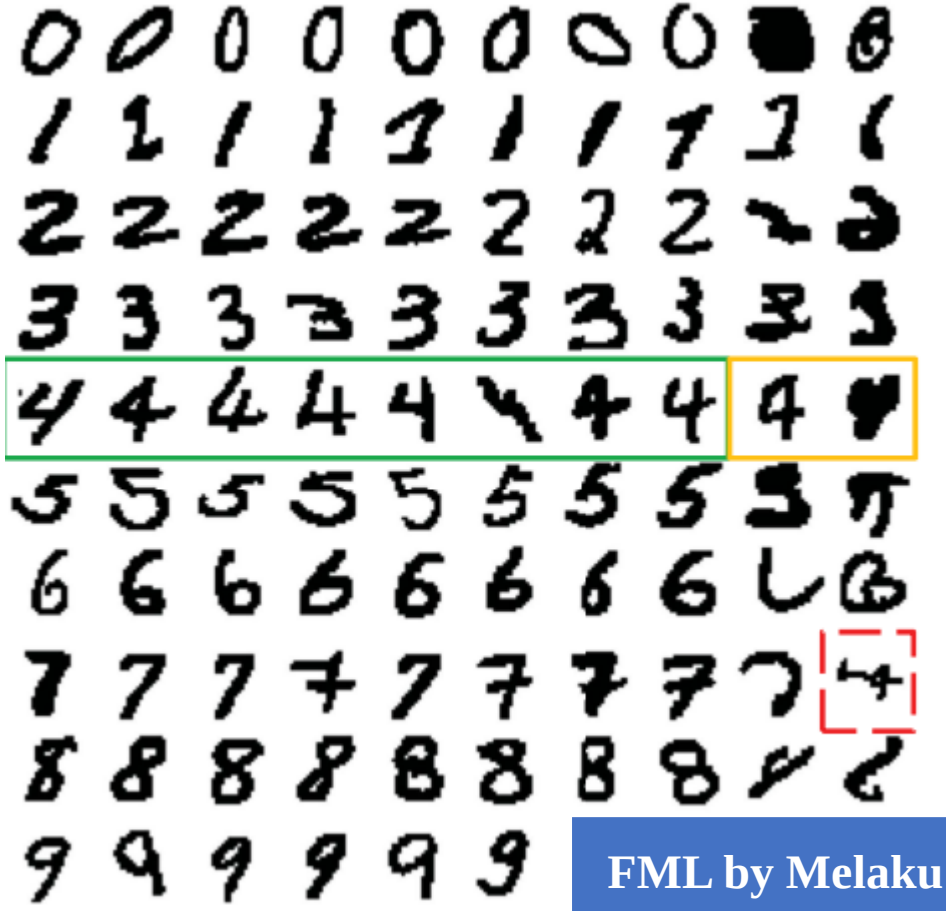
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Supervised Learning: Digit Recognition

What is a ‘2’?



What is a ‘4’?

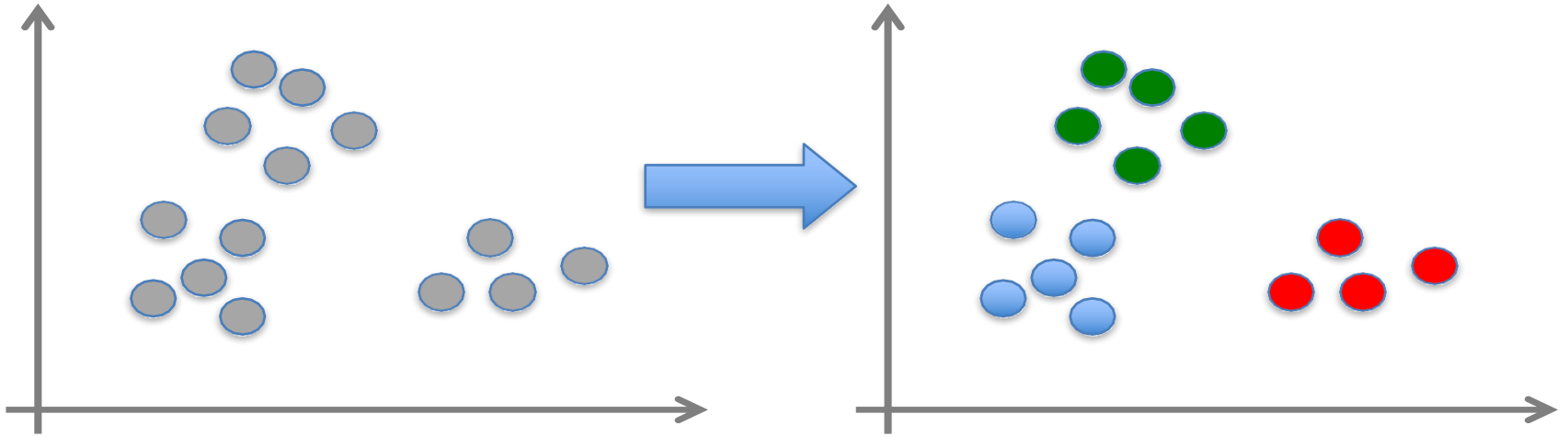


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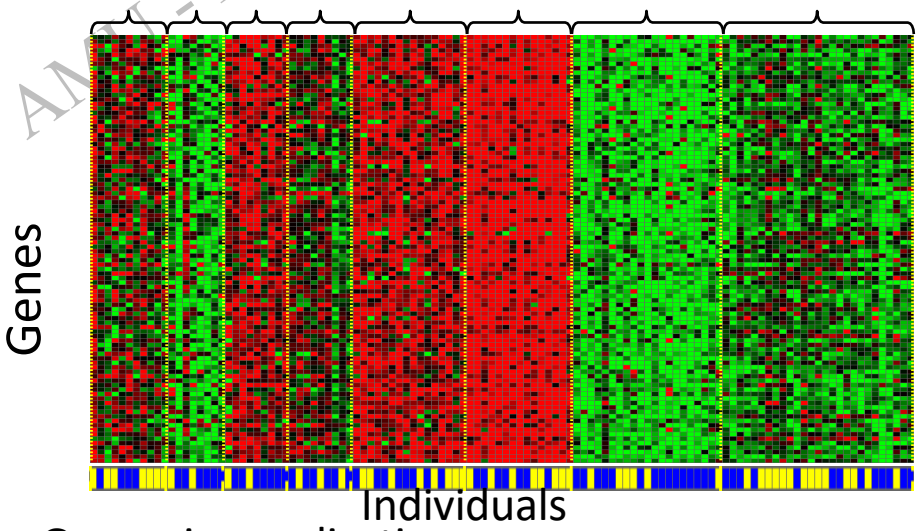
Unsupervised Learning

- The model is trained on **unlabeled dataset**. The model aims to discover hidden patterns, structures, or relationships within the data. Given x_1, x_2, ..., x_n (without labels)
- Output **hidden structure** behind the *x*'s
  - E.g., clustering



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Unsupervised Learning

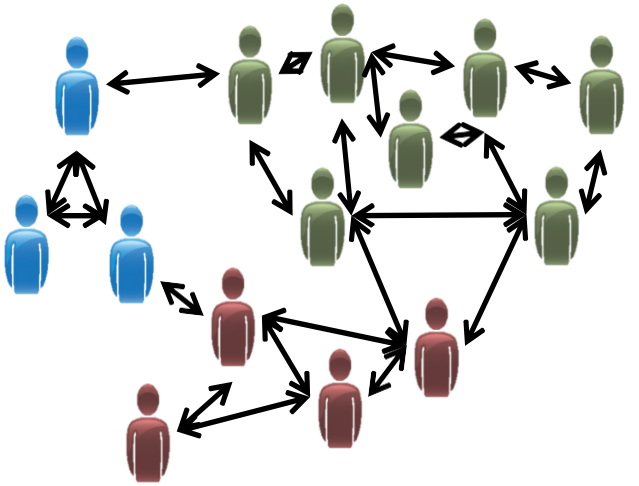


Genomics application:

group individuals by genetic similarity



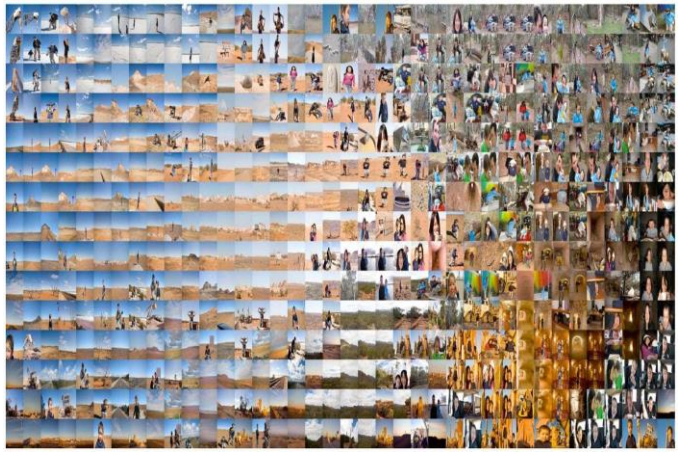
Customer Segmentation



Social network analysis



Astronomical data analysis



Finding image similarity

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Unsupervised Learning

- Independent component analysis – separate a combined signal into its original sources

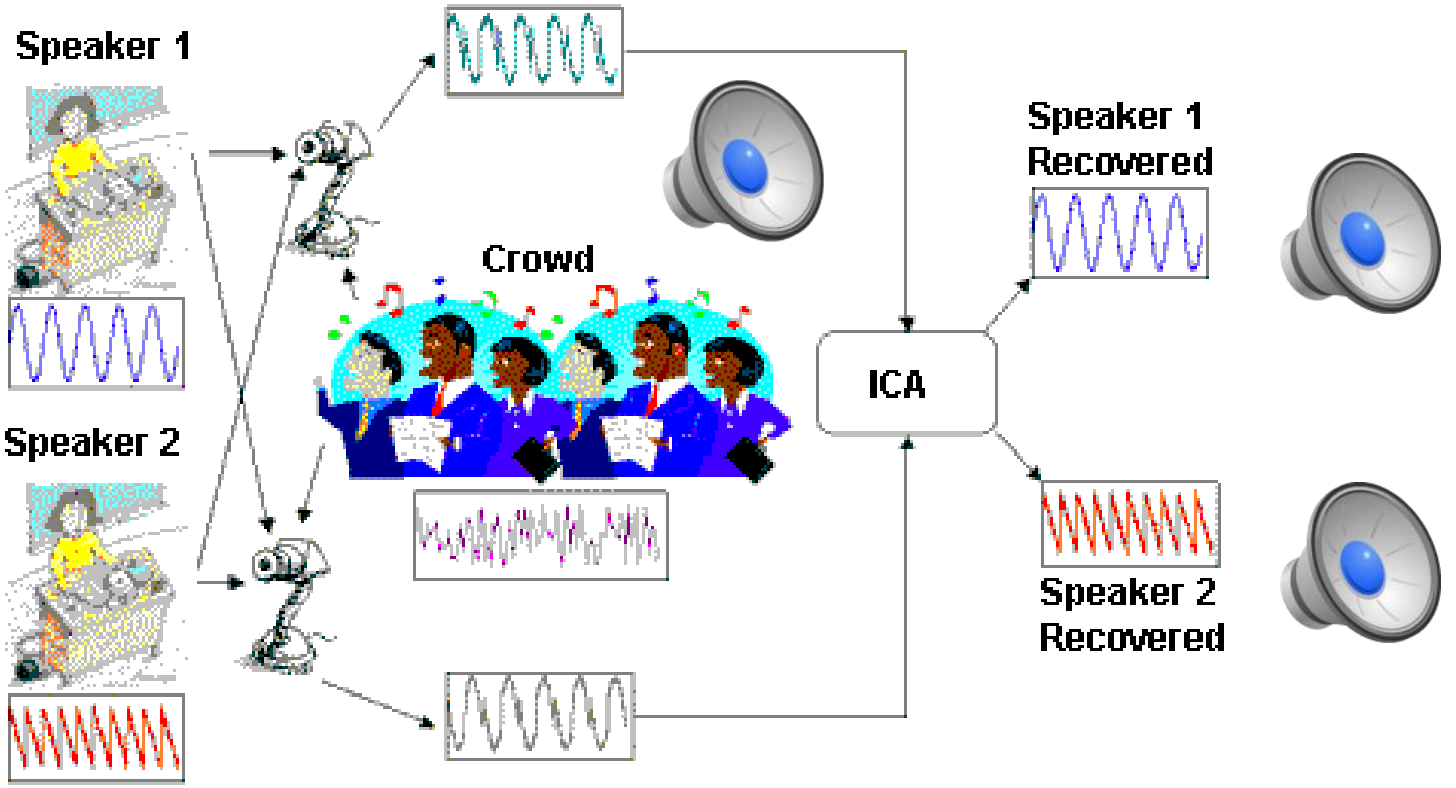


Image credit: statsoft.com Audio from http://www.ism.ac.jp/~shiro/research/blindsep.html

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Unsupervised Learning

- Independent component analysis – separate a combined signal into its original sources

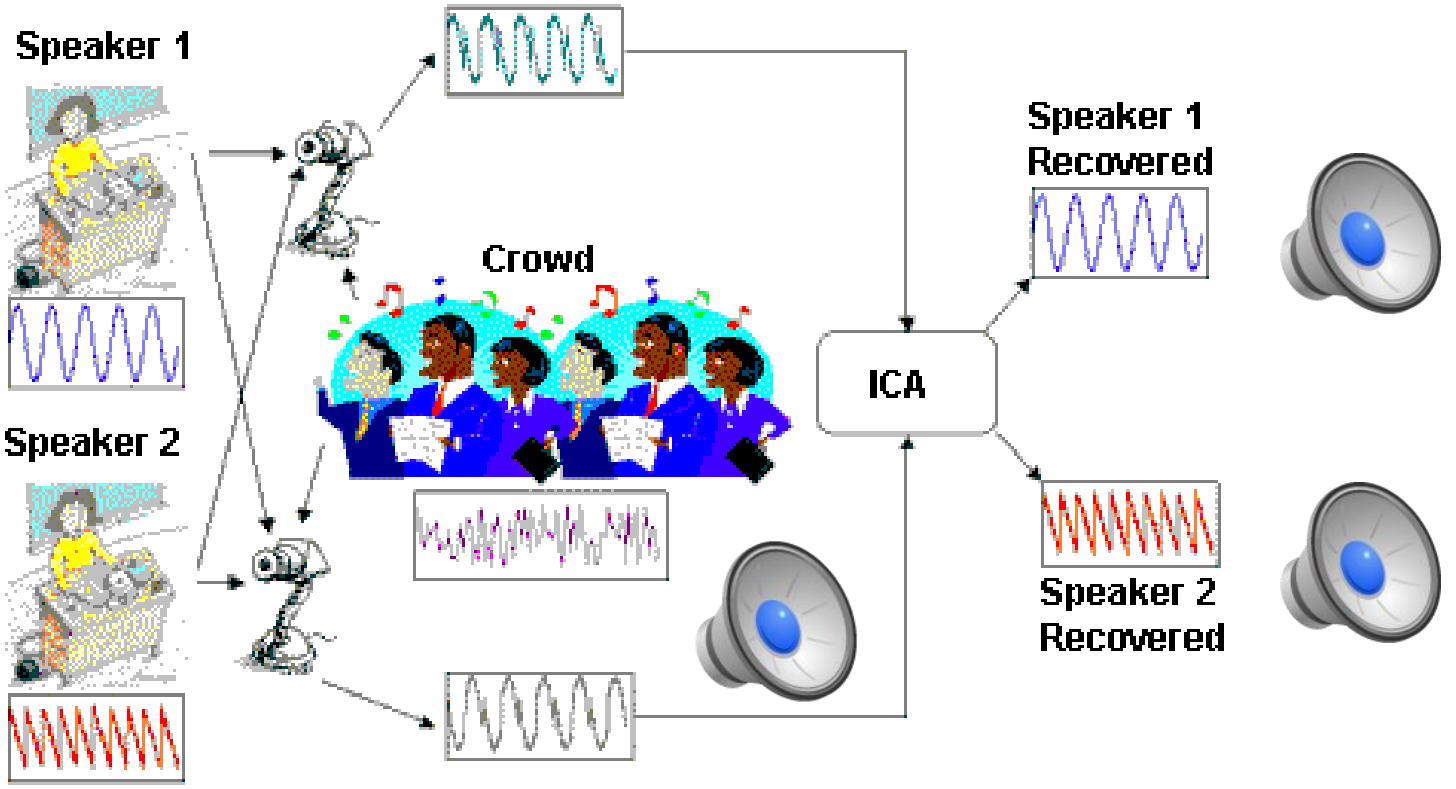


Image credit: statsoft.com Audio from http://www.ism.ac.jp/~shiro/research/blindsep.html

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Semi-supervised Learning:

❖**Definition**: Combines a small amount of **labeled data** with large amounts of **unlabeled data**.

❖This approach combines aspects of supervised and unsupervised learning.

- ✓ It uses both labeled and unlabeled data for training.
- ✓ This can be useful when labeling data is expensive/costly or time-consuming (e.g., medical imaging, speech recognition)..

❖ **Common Applications:**

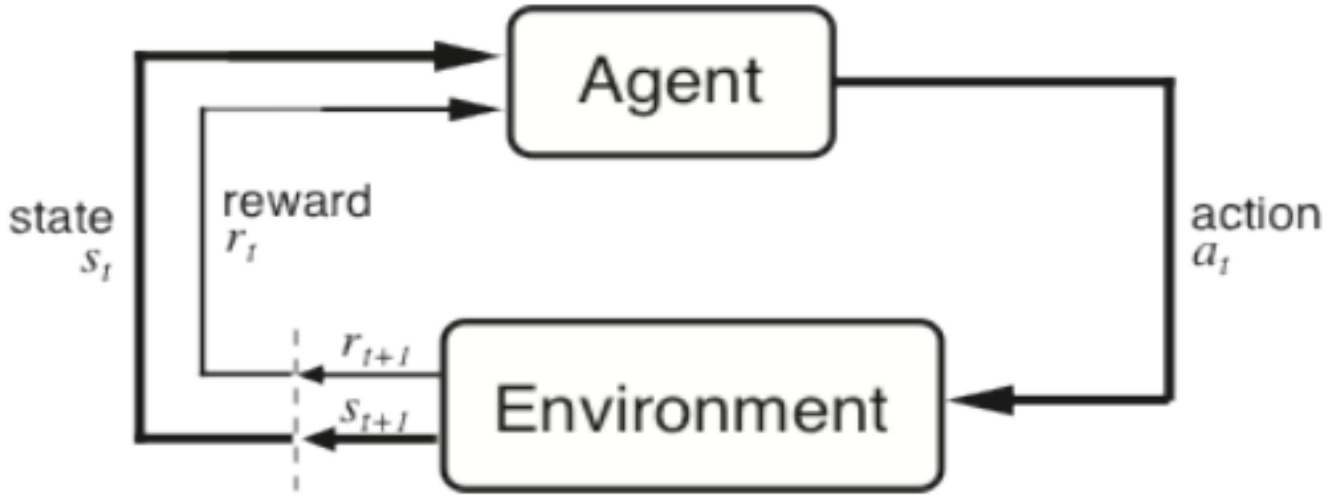
- ✓ Medical imaging (e.g., labeling diseases in X-rays with limited labeled data).
- ✓ Speech recognition (e.g., learning from a few transcribed audio cl

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Reinforcement Learning

- **Reinforcement learning (RL):** involves learning through **interaction with an environment**. An agent learns to take actions in an environment to maximize a reward signal.
- The agent learns through trial and error, receiving rewards or penalties for its actions. The goal is to maximize the cumulative reward.

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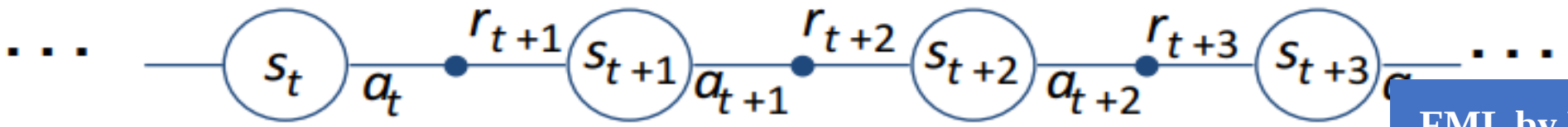
Agent and environment interact at discrete time steps : $t = 0, 1, 2, \mathrm{K}$

Agent observes state at step t: $s_t \in S$

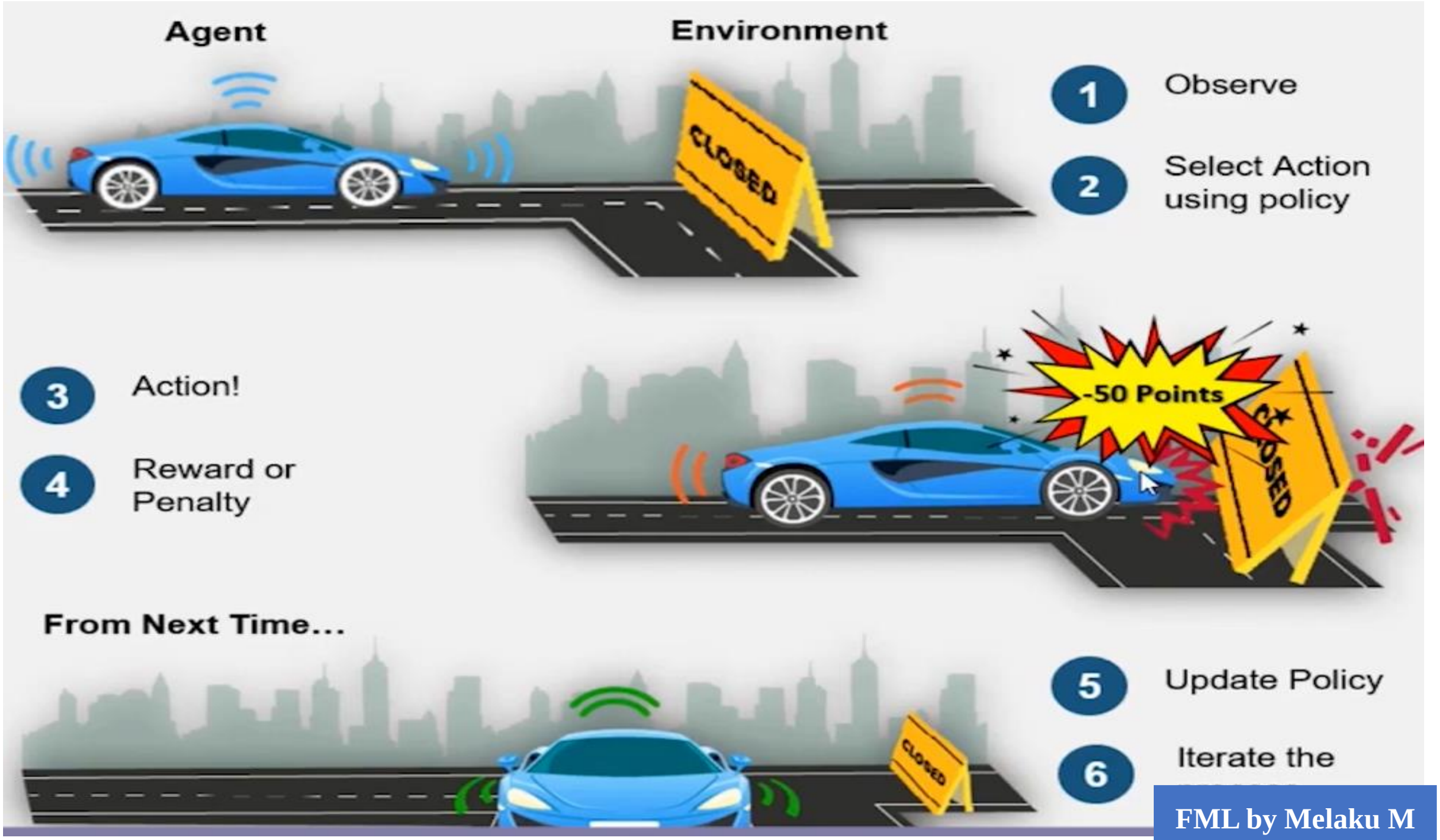
produces action at step t: $a_t \in A(s_t)$

gets resulting reward : $r_{t+1} \in \Re$

and resulting next state : s_{t+1}



FML by Melaku M



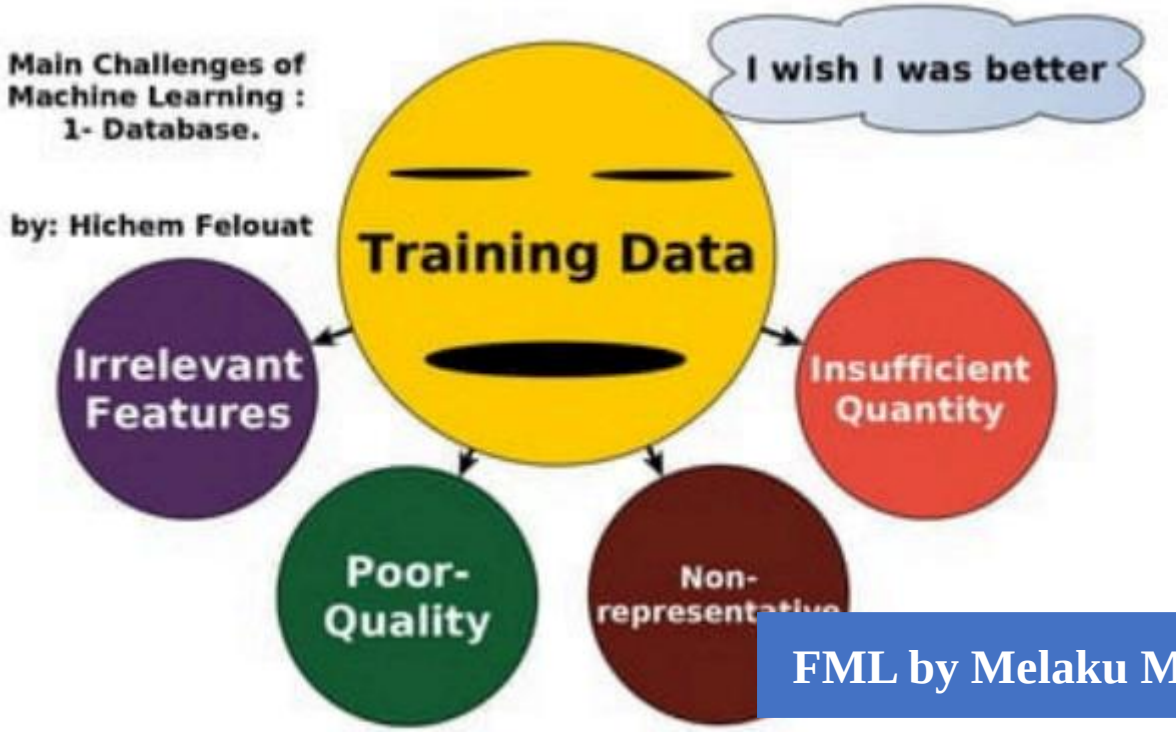
FML by Melaku M

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Main Challenges of Machine Learning

•In short, since your main task is to select a learning algorithm and train it on some data, the two things that can go wrong are 1) **"bad data"** and 2) **"bad algorithm"**.

1. Dataset(training data)



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Main Challenges of Machine Learning - Dataset

1- Insufficient Quantity of Training Data :

•Machine Learning takes a lot of data for most Machine Learning algorithms to work properly. Even for very simple problems you typically need thousands of examples, and for complex problems such as image or speech recognition you may need millions of examples (unless you can reuse parts of an existing model).

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Main Challenges of Machine Learning - Dataset

2) Non-representative Training Data:

•In order to generalize well, it is crucial that your training data be representative of the new cases you want to generalize to. This is true whether you use instance based learning or model-based learning.

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Main Challenges of Machine Learning - Dataset

3) **Poor-Quality Data:**

•If your training data is full of errors, outliers, and noise (e.g., due to poor quality measurements), it will make it harder for the system to detect the underlying patterns, so your system is less likely to perform well. It is often well worth the effort to spend time cleaning up your training data. The truth is, most data scientists spend a significant part of their time doing just that.

Main Challenges of Machine Learning - Dataset

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4 Irrelevant Features:

•Your system will only be capable of learning if the training data contains enough relevant features and not too many irrelevant ones. A critical part of the success of a Machine Learning project is coming up with a good set of features to train on.

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Main Challenges of Machine Learning - Algorithm

1) Overfitting the Training Data:

- Overfitting happens when a model learns the detail and noise in the training **data** to the extent that it negatively impacts the performance of the model on new data.
- ***The model performs well on the training data, but it does not generalize well.***

Overfitting: we have a small training error but large generalization error

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Main Challenges of Machine Learning - Algorithm

2) Underfitting the Training Data:

✓ Underfitting is the opposite of overfitting: it occurs when **your model is too simple** to learn the underlying structure of the data.

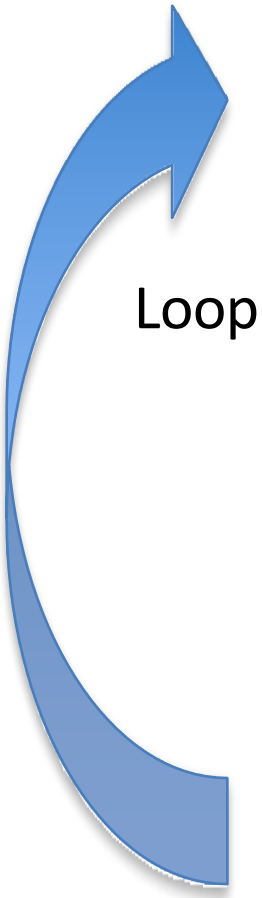
Other ML challenges include:

- **Interpretability**: Some models (e.g., deep learning) are often considered "black boxes," making their decisions difficult to explain.
- **Computational Resources**: Training complex models can require significant time and resources.
- **Bias and Fairness**: Models can inherit biases present in the data.

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ML in Practice

Designing a Learning System



- Understand domain, prior knowledge, and goals
  - Choose the training experience and what is to be learned
    - i.e. the ***target function***
- Data integration, selection, cleaning, pre-processing, etc.
- Learn models
  - Choose a learning algorithm to infer the target function from the experience
- Interpret results
- Consolidate and deploy discovered knowledge



FML by Melaku M